Parth N. Kheni

pkheni@bu.edu | **908-335-5448** | github.com/parthnkheni | linkedin.com/in/parthnkheni

EDUCATION

Boston University **Boston, MA**

B.S. Computer Engineering, Concentration in Machine Learning *Expected Dec 2026*

- Related Coursework: *Machine Learning, Probability Statistics & Data Science, Computation Linear Algebra, Applied Algorithms, Cloud Computing, Software Engineering, Computer Architecture, Differential Equations*

Rutgers University **New Brunswick, NJ**

B.S. Computer Engineering | GPA: 3.97 | Dean's List *Sept 2023 – May 2024*

SKILLS

Programming: Python, SQL, MATLAB, C++ (primary), Java, C#, Objective-C, OOP
Libraries/Frameworks: NumPy, Matplotlib, Pandas, scikit-learn, TensorFlow, Retrofit/Gson, Android Jetpack
Cloud/Tools: Azure AD, GitHub, Microsoft 365, Arduino, Android Studio, API, Power BI, WorkManager
Machine Learning: Predictive Modeling, Statistical Modeling, Time-Series Analysis, Anomaly Detection

EXPERIENCE

Beth Israel Deaconess Medical Center (BIDMC), *Machine Learning Intern* *Sept 2025 – Present*

- Build a GPU-enabled surgical robotics simulation workflow for the da Vinci Research Kit (dVRK, a research platform based on the da Vinci surgical robot) using ROS and Gazebo/AMBF on BU's Shared Computing Cluster (SCC) by containerizing dependencies with Singularity/Apptainer and creating reliable scripts to launch and view simulations remotely.
- Prototype incision-closure tasks in NVIDIA Isaac Sim/ORBIT-Surgical by setting up a reproducible Omniverse/Isaac environment (fixed versions and dependencies) and organizing the required simulation assets for surgical manipulation.
- Improve ML autonomy training reliability by diagnosing a preprocessing mismatch that reduced performance from ~90% to ~30%, curating/labeling demonstration data (e.g. ~80 JIGSAWS trajectories), and collaborating with Johns Hopkins to acquire new dVRK kinematic demos to standardize multimodal inputs (vision + kinematics).

Boston University Center for Space Physics, *Machine Learning Researcher* *Apr 2025 – Present*

- Build end-to-end Python pipelines for POES and OMNI time-series data, generating model-ready datasets with 7M+ 2-second satellite samples per month by ingesting NetCDF files, interpolating gaps, enforcing L-shell (3–9) and MLT quality cuts, and joining with 4-hour windows of 1-minute solar-wind drivers.
- Improve E4_0/E4_90 energetic electron precipitation forecasts, lowering test MAE/RMSE and boosting correlation vs prior baselines, by training Keras neural nets and Ridge regressors with standardized features and time-ordered train/val/test splits.
- Increase interpretability of space-weather predictions across different storm conditions and locations in the radiation belts, improving generalization during rare, high-flux events, by testing different time resolutions (2-second vs 1-minute data), quantile-based flux aggregations, and simple features that capture local time around Earth.

Blue Leaf Technologies, *IT Consultant* *Jul 2023 – Aug 2023*

- Improved employee sign-in and credential flow (25% faster logins, 40% fewer credential bottlenecks, 50% less manual tracking) by deploying Azure AD + Windows Hello for Business, implementing Cloud Kerberos Trust with domain controllers, and automating monitoring/data collection with Python.

PUBLICATIONS & PRESENTATIONS

- Capannolo, L., Pettit, J., Elliott, S., Morales, J. P. B., Bibeau, A., **Kheni, P. N.** *Feb 2026*
  "Monitoring radiation belt electrons at LEO: preliminary results of MAPLE (MAPs of LEO Electrons) model." *COSPAR 2026* – Abstract (submitted)

PROJECTS

LLM-Assisted Incident Summarization & Clustering (Microsoft-affiliated), *Applied ML Engineer* *Jan 2026 – Present*

- Build an end-to-end incident intelligence pipeline: ingest large-scale system/application logs, generate structured incident summaries with LLMs, embed summaries, and cluster incidents to identify duplicates and recurring failure patterns.
- Implement a backend service and lightweight UI/API to explore incidents and clusters, making practical accuracy vs cost/latency tradeoffs to support scalable operation on noisy real-world logs.

Whack-a-Mole Reaction Game on FPGA (Nexys4 DDR, Verilog), *FPGA Design Engineer* *Sept 2025 – Dec 2025*

- Built a real-time Whack-a-Mole game on a Nexys4 DDR FPGA with 30-second multi-difficulty rounds, 5 LED "moles", and per-mole switch/hammer input, using Verilog for debounced I/O, timing, random selection, and top-level integration.
- Improved reliability and integration, delivering a glitch-free demo by owning input and game-control modules, coordinating interfaces across a 4-person team, and iterating with Vivado simulation, timing analysis, and on-board testing.

Personal Finance App, *Android Developer* *Jan 2025 – May 2025*

- Built an Android personal finance app in Android Studio (Java) using Retrofit/Gson + Finnhub.io for real-time quotes (RecyclerView, MPAndroidChart); added SharedPreferences portfolio perisistence, MainActivity →BrowseStocksActivity navigation, and WorkManager + Notifications for price alerts.

AWARDS & HONORS

NSF ACCESS / Jetstream2 Allocation (CIS260066) – $31,444 awarded; 146,000 GPU SUs; 4 TB storage. *Jan 2026 – Present*

GEM Workshop – Invited Speaker (Portland, ME) – selected to present machine learning research. *July 2026*

ACTIVITIES

BU Baja Society of Automotive Engineers, *Head Engineer of Computer Systems* *Sept 2024 – Present*

BU Jalwa Dance Troupe, *Secretary* *Sept 2024 – Present*